



INTERVIEW PREPARATION HANDBOOK

WordPress Developer

Complete Guide

PHP & MySQL fundamentals se le kar Plugin Development, REST API, JWT Authentication, WooCommerce Customization, Directory Websites, Elementor aur Performance Optimization tak — sab kuch Roman Urdu mein, real code examples aur practice tests ke sath.

PHP & OOP

MySQL

WordPress Core

Plugin & REST API

WooCommerce

Elementor

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10 Chapters · 45+ Practice Questions

Full Answer Key & Mock Test

Edition 2.0

WordPress Developer

Complete Guide

PHP, MySQL, WordPress Core, Plugin Development, REST API & JWT, Directory Websites, WooCommerce, SEO & Performance, Elementor, aur Debugging — sab kuch ek jagah, Roman Urdu mein, interview ke liye ready format mein.

AUTHOR	Noman Nadeem WordPress Developer · Elementor Specialist · Figma to WordPress
FORMAT	10 Chapters · 45+ Practice Questions · Complete Answer Key · Full Mock Test
LANGUAGE	Roman Urdu (concepts) + English (code & terminology)
LEVEL	Beginner se Intermediate/Advanced tak — sequence mein parhne ke liye design kiya gaya hai
USE CASE	Interview preparation, revision, aur daily reference guide

© Noman Nadeem. Ye guide personal learning aur interview preparation ke liye tayyar ki gayi hai. Code examples WordPress, WooCommerce, ACF aur Elementor ke public APIs par based hain — apne project mein use karne se pehle latest official documentation zaroor check kar lena.

Is Guide Ko Kaise Use Karein

Ye kitaab sirf parhne ke liye nahi — likh kar aur bol kar practice karne ke liye hai. Har chapter ek hi consistent structure follow karta hai, taake kisi bhi topic par jaldi revise kar sako.

1 Concept

Har cheez Roman Urdu mein simple lafzon mein samjhayi gayi hai, bina jargon ke.

2 Practical Example

Real code jo interview mein likh ke dikha sako. Sirf yaad mat karo — samjho ke har line kyun likhi gayi hai.

3 Problem Solving Exercise

Pehle khud try karo, phir answer dekho — asli seekhna wahin hota hai.

Box Types — Kya Kis Cheez Ke Liye Hai

CODE

Copy-paste ready example. Comments code ke andar hi Roman Urdu mein hain.

INTERVIEW QUESTION

Ye sawal actual interviews mein aata hai — iska jawab bol kar practice karo.

PRACTICE EXERCISE

Khud solve karne wala task. Pehle khud likho, phir aage barho.

ANSWER

Exercise ka solution. Compare karo ke tumhara approach kitna match kiya.

CHAPTER TEST

Chapter khatam hone par khud ko test karo. Answers Answer Key section mein hain.

Suggested Study Plan

- **Din 1-2:** Chapter 1 & 2 (PHP + MySQL) — ye foundation hai, isko skip mat karna.
- **Din 3-5:** Chapter 3 & 4 (WordPress Core + Plugin/REST API) — sabse zyada sawal yahin se aate hain.
- **Din 6-7:** Chapter 5 & 6 (Directory Sites + WooCommerce) — apne projects se relate karke parho.
- **Din 8-10:** Chapter 7 se 10, phir Full Mock Test aur Quick Revision Checklist.

SABSE IMPORTANT TIP

Answers ko ratta mat lagao — har jawab apne alfaz mein **bol kar** practice karo. Jahan apne project ka example de sakte ho, zaroor do.

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PHP Core Concepts

WordPress PHP par bana hua hai — isliye har WordPress interview ki shuruaat PHP se hoti hai. Is chapter mein wo core concepts hain jinke bina baaki sab kuch adhoora hai: variables, arrays, functions, aur sabse important — OOP.

IS CHAPTER MEIN

- Variables & Data Types
- Indexed vs Associative Arrays
- Functions & Variable Scope
- OOP: Class, Inheritance, Encapsulation
- Interfaces & Abstract Classes
- Error Handling (try / catch)

1.1 Variables & Data Types

PHP mein variable hamesha `$` se start hota hai. Type declare nahi karna padta — PHP khud value dekh kar samajh leta hai. Isi ko **loosely typed** kehte hain.

```
$name    = "Noman";           // string
$sage    = 24;                // integer
$price   = 199.99;            // float
$isActive = true;             // boolean
$skills  = ["PHP", "WordPress"]; // array
```

INTERVIEW QUESTION

"PHP loosely typed language hai, iska matlab kya hai?"

Answer: Matlab variable ka type declare nahi karna padta, PHP runtime par value dekh kar khud decide karta hai. `$x = "5";` aur `$x = 5;` dono valid hain, koi error nahi aayega.

1.2 Arrays (Indexed vs Associative)

Indexed array mein data number (0, 1, 2...) se access hota hai, associative array mein apne diye hue key se.

```
// Indexed array
$fruits = ["apple", "banana", "mango"];
echo $fruits[0];           // apple

// Associative array
$product = [
    "name"    => "Beef Steak",
    "price"   => 1500,
    "weight"  => "1kg"
];
echo $product["name"];     // Beef Steak

// Loop through associative array
foreach ($product as $key => $value) {
    echo "$key: $value\n";
}
```

PRACTICE EXERCISE

Ek array banao jisme 5 WooCommerce products hon (name aur price). Loop chala kar sirf wo products print karo jinki price 1000 se zyada hai.

ANSWER

```
$products = [
    ["name" => "Steak", "price" => 1500],
    ["name" => "Mince", "price" => 800],
    ["name" => "Chicken", "price" => 1200],
];

foreach ($products as $product) {
    if ($product["price"] > 1000) {
        echo $product["name"] . "\n";
    }
}
```

1.3 Functions & Scope

```
function calculateDiscount($price, $percent = 10) {
    return $price - ($price * $percent / 100);
}

echo calculateDiscount(1000);           // 900
echo calculateDiscount(1000, 20);      // 800
```


Scope ka concept: Function ke andar declare kiya gaya variable bahar accessible nahi hota, aur bahar ka variable andar accessible nahi hota — jab tak `global` keyword use na karo ya parameter se pass na karo.

```
$x = 10;

function test() {
    echo $x;    // Error / undefined – $x is function ke scope mein hai hi nahi
}
```

TRAP QUESTION

Ye exact code interview mein "output kya aayega?" ke naam se poocha jata hai. Jawab: notice/warning aayegi aur kuch print nahi hoga, kyunki `$x` function ke andar exist hi nahi karta.

1.4 OOP — Sabse Zyada Poocha Jane Wala Topic

Object-Oriented Programming ke bina modern WordPress development (plugins, WooCommerce, Elementor widgets) samajh nahi aati. Neeche wala example weight-based pricing ka real logic hai.

```
class Product {
    protected $name;
    protected $price;

    public function __construct($name, $price) {
        $this->name = $name;
        $this->price = $price;
    }

    public function getPrice() {
        return $this->price;
    }
}

class WeightBasedProduct extends Product {
    private $weight;

    public function __construct($name, $price, $weight) {
        parent::__construct($name, $price);
        $this->weight = $weight;
    }

    public function getPrice() {
        return $this->price * $this->weight;
    }
}

$item = new WeightBasedProduct("Beef", 500, 2); // 500 per kg, 2kg
echo $item->getPrice();                          // 1000
```

INTERVIEW QUESTIONS

"Inheritance kya hai?"

Answer: Ek class doosri class ki properties aur methods use kar sakti hai `extends` keyword ke through — code dobara likhne ki zaroorat nahi padti.

"Encapsulation kya hai?"

Answer: `private` / `protected` laga kar data ko bahar se directly change hone se rokna, taake sirf class ke apne methods hi usko control karein.

"`parent::__construct()` kyun likha?"

Answer: Parent class ka constructor call karne ke liye, taake `$name` aur `$price` bhi properly set ho jayein — warna child class sirf apna `$weight` set karti aur baaki khaali reh jate.

PROJECT SE CONNECTION

Ye directly weight-based pricing wale kaam se related hai — jaise meat/grocery store mein 500/kg wala product 2kg select karne par 1000 ka ho jata hai. Chapter 6 mein isi logic ko WooCommerce hook ke sath dekhoge.

1.5 Interfaces & Abstract Classes

```
interface PaymentGateway {
    public function charge($amount);
}

class StripePayment implements PaymentGateway {
    public function charge($amount) {
        // Stripe API call here
        return "Charged $amount via Stripe";
    }
}
```

Simple explanation: Interface ek "contract" hai — kehta hai ke jo class ye interface implement karegi, usko ye methods zaroor banane honge. Isi wajah se alag alag payment gateways (Stripe, PayPal, COD) ek jaisa structure follow karte hain aur code interchangeable rehta hai.

Abstract class ka farq: Interface sirf method ke naam batata hai (koi code nahi). Abstract class kuch methods ka ready code de sakti hai aur kuch child class ke liye chhod sakti hai.

1.6 Error Handling

```
try {
    if ($stock <= 0) {
        throw new Exception("Product out of stock");
    }
    // process order
} catch (Exception $e) {
    echo "Error: " . $e->getMessage();
}
```

Fayda ye hai ke error aane par poora script crash nahi hota — tum controlled tareeke se message dikha sakte ho ya log kar sakte ho.

Chapter 1 Test

PHP CORE

1. PHP is a _____ typed language.
a) Strictly b) Loosely c) Not typed
2. `final` keyword class ke sath kis kaam aata hai?
3. Ye code kya print karega?

```
$x = "5";
$y = 5;
echo ($x == $y) ? "same" : "different";
```

4. Encapsulation ek line mein example ke sath samjhao.
5. Interface aur Abstract class mein farq kya hai?

Answers: Answer Key section mein (Chapter 1 – PHP).

NOTES & PRACTICE SPACE

MySQL & Database Concepts

WordPress ka poora data database mein hota hai. Agar tum JOIN, index aur `$wpdb` confidently explain kar sakte ho, to tum us 80% candidates se aage nikal jaate ho jo sirf plugin install karna jaante hain.

IS CHAPTER MEIN

- Table structure, Primary & Foreign Key
- SELECT, WHERE, JOIN, GROUP BY
- Indexes aur unka trade-off
- `$wpdb` aur `prepare()` se safe queries

2.1 Basic Table Structure

Ek WooCommerce jaisa simplified products table:

```
CREATE TABLE products (  
  id          INT AUTO_INCREMENT PRIMARY KEY,  
  name        VARCHAR(255) NOT NULL,  
  price       DECIMAL(10,2) NOT NULL,  
  category_id INT,  
  created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

- **Primary Key** — har row ka unique identifier.
- **Foreign Key** — doosri table se relation banata hai (jaise `category_id` categories table se link hota hai).

2.2 Common Queries

```
-- Sab products lao jo 1000 se mehngae hon
SELECT * FROM products WHERE price > 1000;

-- Category ke naam ke sath products lao (JOIN)
SELECT products.name, categories.name AS category
FROM products
JOIN categories ON products.category_id = categories.id;

-- Har category mein kitne products hain (GROUP BY)
SELECT category_id, COUNT(*) AS total
FROM products
GROUP BY category_id;
```

INTERVIEW QUESTION

"INNER JOIN aur LEFT JOIN mein farq?"

Answer: INNER JOIN sirf wo rows deta hai jinka dono tables mein match ho. LEFT JOIN left table ki saari rows deta hai — chahe right table mein match ho ya na ho (na milne par NULL aata hai).

2.3 Indexes

Index ek "shortcut" hai database ko fast search karne ke liye — bilkul kitaab ke index page ki tarah. Jo columns `WHERE` ya `JOIN` mein baar baar use hote hain, un par index lagate hain.

```
CREATE INDEX idx_category ON products(category_id);
```

TRADE-OFF (YAAD RAKHNA)

Index se SELECT fast hota hai, lekin INSERT/UPDATE thora slow ho jata hai — kyunki har naye record ke sath index bhi update karna padta hai.

2.4 WordPress mein Database Access — \$wpdb

```
global $wpdb;

$results = $wpdb->get_results(
    $wpdb->prepare(
        "SELECT * FROM {$wpdb->prefix}posts
        WHERE post_status = %s AND post_type = %s",
        'publish',
        'product'
    )
);
```

INTERVIEW QUESTION

"`$wpdb->prepare()` kyun use karte hain?"

Answer: SQL Injection se bachne ke liye. User se aane wale input ko seedha query mein daalna khatarnaak hota hai — `prepare()` us input ko safely escape karke query mein daalta hai.

PRACTICE EXERCISE

Ek query likho jo WordPress database se wo sab posts laye jo `directory-listing` custom post type ke hain aur last 30 din mein publish hue hain.

ANSWER

```
global $wpdb;

$results = $wpdb->get_results(
    $wpdb->prepare(
        "SELECT * FROM {$wpdb->prefix}posts
        WHERE post_type = %s AND post_status = 'publish'
        AND post_date >= DATE_SUB(NOW(), INTERVAL 30 DAY)",
        'directory-listing'
    )
);
```

1. INNER JOIN aur LEFT JOIN mein farq kya hai?
2. Primary Key aur Foreign Key mein farq batao.
3. Query likho: `orders` table se wo sab orders lao jinka total 5000 se zyada hai, newest first sorted.
4. `$wpdb->prepare()` kyun zaroori hai?
5. Index lagane ka trade-off kya hai?

Answers: Answer Key section mein (Chapter 2 – MySQL).

NOTES & PRACTICE SPACE

WordPress Fundamentals

Ye kitaab ka sabse bara aur sabse important chapter hai. Hooks, Custom Post Types, The Loop, theme structure aur ACF — inhi cheezon se WordPress developer ki asli value banti hai.

IS CHAPTER MEIN

- Actions vs Filters
- Custom Post Types & Taxonomies
- The Loop
- Custom Fields / ACF basics
- Theme file structure
- functions.php & theme setup
- Navigation menus
- Enqueue scripts & styles
- Widgets & sidebars
- Child themes
- Template hierarchy
- ACF Repeater & Flexible Content

3.1 Hooks — Actions vs Filters

Ye WordPress ka sabse core concept hai. Har WordPress developer interview mein ye zaroor poocha jata hai.

- **Action** = "jab ye event ho, tab ye kaam karo" — kuch execute karta hai, value return nahi karta.
- **Filter** = "ye data lo, modify karo, wapas do" — hamesha value return karta hai.

```
// Action example: jab naya user register ho, welcome email bhejo
add_action('user_register', 'send_welcome_email');
function send_welcome_email($user_id) {
    $user = get_userdata($user_id);
    wp_mail($user->user_email, "Welcome!", "Thanks for joining.");
}

// Filter example: page title modify karo
add_filter('the_title', 'add_prefix_to_title');
function add_prefix_to_title($title) {
    return "★ " . $title;
}
```


INTERVIEW TRAP QUESTION

"Filter mein `return` kyun zaroori hai?"

Answer: Kyunki filter ka kaam hi data ko modify karke wapas dena hai. Agar return nahi karoge to poora data khatam ho jayega (null return ho jayega) — jaise title hi gayab ho jayega.

3.2 Custom Post Types (CPT)

Directory website ya kisi bhi structured content wali site banane ke liye ye sabse important cheez hai.

```
function register_directory_listing_cpt() {
    register_post_type('directory_listing', [
        'labels' => [
            'name'          => 'Listings',
            'singular_name' => 'Listing'
        ],
        'public'           => true,
        'has_archive'      => true,
        'supports'         => ['title', 'editor', 'thumbnail', 'custom-fields'],
        'rewrite'          => ['slug' => 'listings'],
        'show_in_rest'     => true,    // REST API ke liye zaroori
    ]);
}
add_action('init', 'register_directory_listing_cpt');
```

Custom Taxonomy (directory ke liye categories/tags, jaise "Location" ya "Business Type"):

```
function register_listing_category() {
    register_taxonomy('listing_category', 'directory_listing', [
        'label'           => 'Categories',
        'hierarchical'    => true,
        'show_in_rest'    => true,
    ]);
}
add_action('init', 'register_listing_category');
```

INTERVIEW QUESTION

"Custom Post Type aur regular Page mein farq?"

Answer: CPT apna structured data type hota hai (jaise Listings, Products, Testimonials) jisko alag se manage kiya ja sakta hai — apni taxonomy, apna archive page, apna REST endpoint hota hai. Page sirf static content ke liye hota hai.

3.3 The Loop

The Loop WordPress ka wo mechanism hai jisse posts fetch ho kar ek ek karke display hote hain.

```
<?php if (have_posts()) : while (have_posts()) : the_post(); ?>

    <h2><?php the_title(); ?></h2>
    <p><?php the_excerpt(); ?></p>

<?php endwhile; endif; ?>
```

3.4 Custom Fields / ACF (Directory Sites Ke Liye Critical)

Directory listing ke liye har entry ka apna extra data hota hai — address, phone, rating, timings.

```
// ACF field value nikalna
$phone = get_field('business_phone');
$address = get_field('business_address');

// Native WordPress custom field (bina ACF ke)
$phone = get_post_meta(get_the_ID(), 'business_phone', true);
```

PRACTICE EXERCISE

Directory listing ke liye ek query likho jo sirf "Lahore" location wale listings dikhaye, custom field `city` ke basis par.

ANSWER

```
$args = [
    'post_type' => 'directory_listing',
    'meta_query' => [
        [
            'key' => 'city',
            'value' => 'Lahore',
            'compare' => '='
        ]
    ]
];

$query = new WP_Query($args);
```

3.5 Theme Development — File Structure

Ek WordPress theme minimum in files se banti hai:

```
style.css      -> theme ki info + CSS (header comment zaroori hai)
functions.php  -> theme setup, hooks, features enable karna
index.php      -> fallback template (agar koi specific file na mile)
header.php     -> <head> aur site ka top part
footer.php     -> site ka bottom part
single.php     -> single blog post template
page.php       -> single page template
archive.php    -> category / tag / date listing page
```

`style.css` ke sabse upar ye comment hona zaroori hai — isi se WordPress theme ko pehchanta hai:

```
/*
Theme Name: My Custom Theme
Author: Noman Nadeem
Version: 1.0
*/
```

3.6 functions.php aur Theme Setup

```
function my_theme_setup() {
    add_theme_support('title-tag');
    add_theme_support('post-thumbnails');

    register_nav_menus([
        'primary' => 'Primary Menu',
        'footer'  => 'Footer Menu',
    ]);
}
add_action('after_setup_theme', 'my_theme_setup');
```

3.7 Navigation Menu Kaise Banate Hain

Do steps hote hain. Pehle `functions.php` mein menu location register karo (upar wale code mein already kar diya), phir template file (jaise `header.php`) mein usko display karo:

```
wp_nav_menu([
    'theme_location' => 'primary',
    'container'      => 'nav',
    'menu_class'     => 'main-menu',
]);
```

Uske baad WordPress admin mein **Appearance** → **Menus** se ja kar menu items add karo aur "Primary Menu" location assign karo. Bina location register kiye, admin mein wo menu dikhega hi nahi.

3.8 Enqueue Scripts & Styles

```
function my_theme_scripts() {
    wp_enqueue_style('main-style', get_stylesheet_uri());

    wp_enqueue_script(
        'main-script',
        get_template_directory_uri() . '/js/main.js',
        ['jquery'], // dependencies
        '1.0',      // version (cache busting)
        true        // footer mein load karo
    );
}
add_action('wp_enqueue_scripts', 'my_theme_scripts');
```

INTERVIEW QUESTION

"CSS/JS ko seedha <head> mein <script> tag se load kyun nahi karte?"

Answer: `wp_enqueue_script` / `wp_enqueue_style` conflicts avoid karta hai (same file do baar load na ho), dependencies manage karta hai (jaise jQuery pehle load ho apni script se pehle), aur version number ke through cache busting bhi ho jati hai.

3.9 Widgets & Sidebars

```
function my_theme_widgets_init() {
    register_sidebar([
        'name'          => 'Main Sidebar',
        'id'            => 'sidebar-1',
        'before_widget' => '<div class="widget">',
        'after_widget'  => '</div>',
    ]);
}
add_action('widgets_init', 'my_theme_widgets_init');
```

Template file mein display karna:

```
if (is_active_sidebar('sidebar-1')) {
    dynamic_sidebar('sidebar-1');
}
```

3.10 Child Themes

Client project mein parent/purchased theme ko directly edit karna risky hota hai — agla update sab changes overwrite kar dega. Isliye child theme banate hain.

`style.css` (child theme ke andar):

```
/*
Theme Name: My Theme Child
Template: my-theme
*/
```

`functions.php` (child theme):

```
add_action('wp_enqueue_scripts', function() {
    wp_enqueue_style(
        'parent-style',
        get_template_directory_uri() . '/style.css'
    );
});
```

COMMON MISTAKE

Template: line mein parent theme ka **exact folder name** likhna hota hai, warna site white screen de degi.

3.11 Template Hierarchy

WordPress khud decide karta hai ke konsi file load karni hai, ek fixed priority order ke hisaab se. Single post ke liye order kuch aisa hota hai:

```
single-{post-type}.php → single.php → singular.php → index.php
```

INTERVIEW QUESTION

"Agar ek custom post type 'property' ke liye alag design chahiye to file ka naam kya rakhoge?"

Answer: `single-property.php` — WordPress isko automatically us post type ke single pages ke liye pick kar lega.

3.12 ACF (Advanced Custom Fields) — Deep Dive

Recap: ACF ek plugin hai jo WordPress admin mein custom fields add karne ka UI deta hai, bina khud meta box ka PHP code likhe (basic `get_field()` section 3.4 mein dekh chuke ho).

Repeater Field

Jab ek hi type ka data multiple times chahiye — jaise ek directory listing ke multiple phone numbers:

```
if (have_rows('phone_numbers')) {
    while (have_rows('phone_numbers')) {
        the_row();
        echo get_sub_field('phone_number') . '<br>';
    }
}
```

Flexible Content Field

Jab page ke different sections alag alag layout ke ho sakte hain — page-builder jaisa flexible structure:

```
if (have_rows('page_sections')) {
    while (have_rows('page_sections')) {
        the_row();

        if (get_row_layout() == 'hero_section') {
            echo get_sub_field('hero_title');
        } elseif (get_row_layout() == 'testimonial_section') {
            echo get_sub_field('testimonial_text');
        }
    }
}
```

`acf/save_post` Hook

Jab ACF field save ho, tab custom logic chalane ke liye:

```
add_action('acf/save_post', 'sync_price_after_acf_save', 20);

function sync_price_after_acf_save($post_id) {
    $weight = get_field('weight', $post_id);
    // yahan price recalculate karna, ya kisi doosre system ko sync karna
}
```

"ACF Repeater aur Flexible Content mein farq?"

Chapter 3 Test

1. Action aur Filter mein core farq kya hai?
2. `show_in_rest` custom post type mein kyun set karte ho?
3. Code likho: ek custom taxonomy register karo naam "brand" ke sath, post type "product" ke liye.
4. Custom Post Type aur normal Page mein farq batao.
5. The Loop kis kaam aata hai?
6. Navigation menu register karne aur display karne ke do steps kya hain?
7. `wp_enqueue_script` seedha `<script>` tag se better kyun hai?
8. Child theme kab aur kyun banate hain?
9. ACF Repeater aur Flexible Content field mein farq batao.
10. `acf/save_post` hook kis kaam aata hai?

NOTES & PRACTICE SPACE

Plugin Development & REST API

Yahan se tum "theme lagane wale" se "developer" ban jaate ho. Plugin structure, REST API endpoints, security, custom tables aur JWT authentication — ye wo topics hain jo senior-level interviews mein decide karte hain ke tumhe kaunsi salary offer hogi.

IS CHAPTER MEIN

- Plugin ka basic structure
- REST API endpoint banana
- Apne custom hooks banana
- Shortcodes
- Admin Settings page
- Security: sanitize, escape, nonce
- Custom database table
- Cron jobs
- JWT authentication

4.1 Basic Plugin Structure

```
<?php
/**
 * Plugin Name: My Custom Plugin
 * Version: 1.0
 */

if (!defined('ABSPATH')) exit; // Direct access rokne ke liye

register_activation_hook(__FILE__, 'my_plugin_activate');
function my_plugin_activate() {
    // Database table banao, default options set karo
}

register_deactivation_hook(__FILE__, 'my_plugin_deactivate');
function my_plugin_deactivate() {
    // Cleanup karo
}
```

4.2 REST API Endpoint Banana

Headless setups aur mobile app backends isi par based hote hain — interview mein isko confidently explain karna bara plus point hai.

```
add_action('rest_api_init', function () {
    register_rest_route('myapp/v1', '/products', [
```



```

        'methods'           => 'GET',
        'callback'          => 'get_products_for_app',
        'permission_callback' => '__return_true', // ya JWT check
    ]);
});

function get_products_for_app($request) {
    $args = ['post_type' => 'product', 'posts_per_page' => 20];
    $query = new WP_Query($args);

    $products = [];
    foreach ($query->posts as $post) {
        $products[] = [
            'id'       => $post->ID,
            'name'      => $post->post_title,
            'price'     => get_post_meta($post->ID, '_price', true),
        ];
    }

    return new WP_REST_Response($products, 200);
}

```

INTERVIEW QUESTION

"REST API endpoint secure kaise banate ho?"

Answer: `permission_callback` mein check lagate hain — jaise JWT token verify karna ya nonce check karna, taake sirf authorized user hi data access kar sake.

```

'permission_callback' => function($request) {
    $token = $request->get_header('Authorization');
    return verify_jwt_token($token); // apna verification function
}

```

4.3 Hooks & Filters Custom Plugin Mein

```

// Apna custom action banao
do_action('before_order_placed', $order_data);

// Koi doosra plugin/theme is action ko listen kar sakta hai
add_action('before_order_placed', function($order_data) {
    // extra logic, jaise notification bhejna
});

```

Isi tarah tumhara plugin doosron ke liye **extendable** ban jata hai — yahi cheez professional plugins ko amateur plugins se alag karti hai.

4.4 Shortcodes

```
function my_custom_button_shortcode($atts) {
    $atts = shortcode_atts([
        'text' => 'Click Me',
        'link' => '#'
    ], $atts);

    return '<a href="' . esc_url($atts['link']) . '" class="btn">'
        . esc_html($atts['text']) . '</a>';
}
add_shortcode('custom_button', 'my_custom_button_shortcode');
```

Editor mein use: `[custom_button text="Buy Now" link="/shop"]`

4.5 Admin Settings Page (Settings API)

```
add_action('admin_menu', function() {
    add_options_page(
        'My Plugin Settings', // page title
        'My Plugin',         // menu title
        'manage_options',    // capability
        'my-plugin',         // slug
        'my_plugin_settings_page'
    );
});

function my_plugin_settings_page() { ?>
    <div class="wrap">
        <h1>My Plugin Settings</h1>
        <form method="post" action="options.php">
            <?php
                settings_fields('my_plugin_options');
                do_settings_sections('my-plugin');
                submit_button();
            ?>
        </form>
    </div>
    <?php }
```

Simple explanation: `add_options_page()` se WordPress admin ke Settings menu mein apna page add hota hai. Settings API forms ko secure aur consistent tareeke se save karta hai.

4.6 Security Basics

```
// Input sanitize karna (user se aane wala data safe banana)
$name = sanitize_text_field($_POST['name']);

// Output escape karna (screen par print karte waqt safe banana)
echo esc_html($name);

// Nonce verify karna (CSRF attack se bachne ke liye)
if (!wp_verify_nonce($_POST['my_nonce'], 'my_action')) {
    die('Security check failed');
}
```

RULE OF THUMB (INTERVIEW MEIN YAHİ BOLNA)

"Sanitize on input, escape on output." Matlab jab data andar aaye to sanitize karo, aur jab bahar print karo to escape karo.

4.7 Custom Database Table

Jab plugin ka data WordPress ke normal posts/postmeta structure mein fit nahi hota (jaise logs, analytics, bookings), tab apni table banate hain:

```
function my_plugin_create_table() {
    global $wpdb;

    $table_name      = $wpdb->prefix . 'my_custom_logs';
    $charset_collate = $wpdb->get_charset_collate();

    $sql = "CREATE TABLE $table_name (
        id            mediumint(9) NOT NULL AUTO_INCREMENT,
        message       text NOT NULL,
        created_at     datetime DEFAULT CURRENT_TIMESTAMP,
        PRIMARY KEY (id)
    ) $charset_collate;";

    require_once ABSPATH . 'wp-admin/includes/upgrade.php';
    dbDelta($sql);
}

register_activation_hook(__FILE__, 'my_plugin_create_table');
```

4.8 Cron Jobs (Scheduled Tasks)

```
add_action('my_daily_cleanup_hook', 'run_daily_cleanup');

if (!wp_next_scheduled('my_daily_cleanup_hook')) {
    wp_schedule_event(time(), 'daily', 'my_daily_cleanup_hook');
}

function run_daily_cleanup() {
    // purane logs delete karna, waghera
}
```

INTERVIEW QUESTION

"WordPress mein cron job real cron se kaise alag hai?"

Answer: WordPress ka cron real server cron nahi hai — ye site par visitor aane par trigger hota hai (isi liye ise "fake cron" bhi kehte hain). Isliye low-traffic site par scheduled task time par nahi chalta jab tak koi visit na kare. Fix: server ka real cron laga kar `wp-cron.php` ko call karwana.

4.9 JWT Authentication — Deep Dive

Definition: JWT (JSON Web Token) ek token-based authentication method hai. Login hone par server ek signed token generate karta hai jisme user ki info encode hoti hai. Client (mobile app) har request ke sath ye token bhejta hai, server usko verify karta hai — bina kisi session ko database ya server memory mein store kiye. Isi wajah se ise **stateless authentication** kehte hain.

Token 3 parts se banta hai, dots se separate:

Header . Payload . Signature

Session ki jagah JWT kyun? Mobile apps stateless REST APIs use karte hain, jahan cookies/sessions maintain karna mushkil hota hai (khaaskar cross-device). JWT stateless hai — server ko kuch store nahi karna padta, sirf signature verify karta hai ke token genuine hai ya nahi.

Implementation

```
define('JWT_SECRET', 'a-long-random-secret-key');

// Login hone par token generate karna
function generate_jwt($user_id) {
    $header = base64_encode(json_encode(['typ' => 'JWT', 'alg' => 'HS256']));
    $payload = base64_encode(json_encode([
        'user_id' => $user_id,
        'exp'     => time() + (7 * 24 * 60 * 60), // 7 din valid
    ]));
}
```

```

    $signature = base64_encode(
        hash_hmac('sha256', "$header.$payload", JWT_SECRET, true)
    );

    return "$header.$payload.$signature";
}

// Har request par token verify karna
function verify_jwt_token($token) {
    $parts = explode('.', $token);
    if (count($parts) !== 3) return false;

    list($header, $payload, $signature) = $parts;

    $expected_signature = base64_encode(
        hash_hmac('sha256', "$header.$payload", JWT_SECRET, true)
    );

    if (!hash_equals($expected_signature, $signature)) {
        return false; // signature match nahi hui – token fake/tampered hai
    }

    $data = json_decode(base64_decode($payload), true);

    if ($data['exp'] < time()) {
        return false; // token expire ho chuka
    }

    return $data['user_id'];
}

```

INTERVIEW QUESTION

"JWT secure kaise hai — koi bhi token forge kyun nahi kar sakta?"

Answer: Signature `JWT_SECRET` (ek private key jo sirf server ke paas hoti hai) se banti hai. Agar client payload change bhi kar de (jaise apna `user_id` badal de), signature match nahi hogi kyunki uske paas secret key nahi hai — isliye tampered token turant reject ho jata hai.

1. Plugin file ke shuru mein `if (!defined('ABSPATH')) exit;` kyun likhte hain?
2. REST API route register karne ke liye kaunsa hook use hota hai?
3. `permission_callback` ka kaam kya hai?
4. Code likho: ek POST method wala REST endpoint `myapp/v1/orders` banao.
5. Custom action `do_action()` aur normal function call mein farq kya hai?
6. Shortcode kaise banate hain — ek example do.
7. "Sanitize on input, escape on output" ka matlab samjhao.
8. WordPress ka cron job real server cron se kaise alag hai?
9. JWT ko "stateless authentication" kyun kehte hain?
10. Koi client JWT token ka data (jaise `user_id`) khud se change karke fake authenticate kyun nahi ho sakta?

Answers: Answer Key section mein (Chapter 4 – Plugin & REST API).

NOTES & PRACTICE SPACE

Directory Website Development

Directory sites (business listings, property portals, job boards) WordPress ka sabse common "custom development" project hain. Inka poora structure sirf teen building blocks par khara hai.

IS CHAPTER MEIN

- Directory site ka core structure
- Search & filter functionality
- Performance problem aur uska fix

5.1 Core Structure

Directory website 3 cheezon par based hoti hai:

1. **Custom Post Type** — Listings
2. **Custom Taxonomy** — Category, Location
3. **Custom Fields** — Phone, Address, Rating, Hours

5.2 Search & Filter Functionality

Taxonomy aur meta field, dono par ek sath filter lagana — yahi directory site ka dil hai.

```
function filter_listings_by_category($category, $location) {
    $args = [
        'post_type' => 'directory_listing',

        'tax_query' => [
            [
                'taxonomy' => 'listing_category',
                'field'     => 'slug',
                'terms'     => $category,
            ]
        ],

        'meta_query' => [
            [
                'key'       => 'city',
                'value'     => $location,
                'compare'   => '='
            ]
        ]
    ];
}
```

```
return new WP_Query($args);
}
```

INTERVIEW QUESTION

"Directory website mein sabse zyada performance issue kahan aata hai?"

Answer: Jab listings hazaron mein ho jaati hain aur har listing ke multiple meta fields par filter/search karna padta hai, to `meta_query` slow ho jati hai kyunki `wp_postmeta` table properly indexed nahi hoti.

Solution: important fields ke liye custom database table banana, ya `WP_Query` ki jagah direct optimized SQL query likhna.

Answer: Jab listings hazaron mein ho jaati hain aur har listing ke multiple meta fields par filter/search karna padta hai, to `meta_query` slow ho jati hai kyunki `wp_postmeta` table properly indexed nahi hoti.

Solution: important fields ke liye custom database table banana, ya `WP_Query` ki jagah direct optimized SQL query likhna.

Chapter 5 Test

DIRECTORY SITES

DIRECTORY SITES

1. Directory website ke 3 core building blocks kya hain?
2. Code likho: ek query jo "Karachi" city aur "restaurant" category — dono conditions se filter kare.
3. Directory sites large scale par slow kyun ho jati hain, aur fix kya hai?

Answers: Answer Key section mein (Chapter 5 – Directory Sites).

NOTES & PRACTICE SPACE

WooCommerce Customization

WooCommerce khud WordPress ke actions/filters system (section 3.1) par bana hai — bas apne custom hook names use karta hai jo specific events (cart update, checkout, order status change) par fire hote hain. Neeche har important hook do parts mein samjhaya gaya hai: **Definition** (kab fire hota hai, kya parameters deta hai) aur **Implementation** (real code jo interview mein likh sako).

IS CHAPTER MEIN — 8 CORE HOOKS

- woocommerce_product_get_price
- woocommerce_before_add_to_cart_button
- woocommerce_add_cart_item_data
- woocommerce_cart_calculate_fees
- woocommerce_after_order_notes
- woocommerce_checkout_process
- woocommerce_order_status_changed
- woocommerce_thankyou

6.1 WooCommerce Hooks Kya Hote Hain

Agar Chapter 3 ka actions vs filters wala concept clear hai, to WooCommerce customization sirf "sahi hook dhoondhna" reh jati hai. Filter hooks se data modify hota hai (price, cart data), action hooks se kuch execute hota hai (field print karna, notification bhejna). Order flow ke hisaab se hooks ka sequence ye hai:

Product page → Add to cart → Cart totals → Checkout fields
→ Checkout validation → Order create → Status change → Thank You

6.2 woocommerce_product_get_price FILTER

Definition: Ye filter tab chalta hai jab bhi WooCommerce kisi product ki price calculate ya display karta hai — cart, single product page, shop listing, sab jagah. Do parameters milte hain: `$price` (current price) aur `$product` (poora product object). Filter hone ki wajah se hamesha modified `$price` return karna zaroori hai.

```
add_filter('woocommerce_product_get_price', 'custom_weight_price', 10, 2);

function custom_weight_price($price, $product) {
    $weight = $product->get_weight();

    if ($weight) {
        return $price * $weight;
    }

    return $price;
}
```

`get_weight()` WooCommerce ka built-in method hai jo product ka weight nikalta hai. Agar weight set hai to price ko us weight se multiply kar dete hain — yani 500/kg wala product agar cart mein 2kg select hota hai to final price 1000 ban jata hai.

6.3 woocommerce_before_add_to_cart_button ACTION

Definition: Single product page par "Add to Cart" button se theek pehle fire hota hai. Koi extra field, note ya custom UI element add karna ho to yahan karte hain. Ye action hai, isliye koi value return nahi karta — sirf output karta hai.

```
add_action('woocommerce_before_add_to_cart_button', 'add_delivery_note_field');

function add_delivery_note_field() {
    echo '<p class="delivery-note">
        <label>Delivery Instructions:</label>
        <input type="text" name="delivery_note" />
    </p>';
}
```

Ye product page par ek text field add karta hai, jo aage cart data ke sath save ho sakta hai.

6.4 woocommerce_add_cart_item_data FILTER

Definition: Jab customer "Add to Cart" click karta hai, cart item ka data banane se pehle ye filter chalta hai. Do parameters: `$cart_item_data` (array jisme extra data daal sakte ho) aur `$product_id`. Return hamesha `$cart_item_data` array hi hona chahiye, warna cart item corrupt ho jayega.

```
add_filter('woocommerce_add_cart_item_data', 'add_custom_cart_data', 10, 2);

function add_custom_cart_data($cart_item_data, $product_id) {
    if (isset($_POST['delivery_note'])) {
        $cart_item_data['delivery_note'] = sanitize_text_field($_POST['delivery_note']);
    }

    return $cart_item_data;
}
```

Upar wale field se aaya `delivery_note` POST data yahan cart item ke sath attach hota hai, taake wo checkout aur order tak carry ho jaye.

6.5 woocommerce_cart_calculate_fees ACTION

Definition: Cart aur checkout ke totals calculate hote waqt fire hota hai. Yahan custom fee (jaise delivery charge, handling fee) add kar sakte hain. Parameter mein `$cart` object milta hai.

```
add_action('woocommerce_cart_calculate_fees', 'add_delivery_fee');

function add_delivery_fee($cart) {
    if (is_admin() && !defined('DOING_AJAX')) return;

    $cart->add_fee('Delivery Charge', 150);
}
```

`is_admin()` check isliye hai taake ye fee admin panel mein dobara add na ho jaye — sirf frontend cart/checkout par chale.

6.6 woocommerce_after_order_notes ACTION

Definition: Checkout page par "Order Notes" field ke turant baad fire hota hai. Custom checkout fields add karne ke liye ye standard jagah hai.

```
add_action('woocommerce_after_order_notes', 'add_delivery_instructions_field');

function add_delivery_instructions_field($checkout) {
    woocommerce_form_field('delivery_instructions', [
        'type' => 'textarea',
        'label' => 'Delivery Instructions',
    ], $checkout->get_value('delivery_instructions'));
}
```

`woocommerce_form_field()` WooCommerce ka helper function hai jo field ko checkout ke design ke sath consistent bana kar render karta hai.

6.7 woocommerce_checkout_process ACTION

Definition: Jab customer "Place Order" click karta hai, order create hone se **pehle** ye action chalta hai. Yahan validation karte hain — agar kuch galat ho to `wc_add_notice()` se error dikha kar checkout rok sakte hain.

```
add_action('woocommerce_checkout_process', 'validate_delivery_instructions');

function validate_delivery_instructions() {
    if (empty($_POST['delivery_instructions'])) {
        wc_add_notice('Please add delivery instructions.', 'error');
    }
}
```

6.8 woocommerce_order_status_changed ACTION

Definition: Jab bhi order ka status change hota hai (jaise processing se completed), ye action chalta hai. Teen parameters milte hain: `$order_id`, `$old_status`, `$new_status`.

```
add_action('woocommerce_order_status_changed', 'notify_rider_on_status_change', 10, 3);

function notify_rider_on_status_change($order_id, $old_status, $new_status) {
    if ($new_status === 'processing') {
        // rider ko notification bhejna, jaise Firebase push
    }
}
```

Yahi pattern rider/delivery tracking systems mein use hota hai — jab order "processing" mein jata hai, rider ko app par notification chali jati hai.

6.9 woocommerce_thankyou ACTION

Definition: Order complete hone ke baad "Thank You" page load hote waqt fire hota hai. `$order_id` parameter milta hai. Order confirmation ke baad koi extra kaam karna ho (jaise external system ko notify karna) to yahan karte hain.

```
add_action('woocommerce_thankyou', 'send_order_to_mobile_app', 10, 1);

function send_order_to_mobile_app($order_id) {
    $order = wc_get_order($order_id);
    // order data ko apne REST API / mobile app ko push karna
}
```

6.10 Common Interview Question

INTERVIEW QUESTION

"WooCommerce mein custom pricing logic kaise implement karoge, jaise weight ke hisaab se price?"

Answer: `woocommerce_product_get_price` filter use karke — product ke weight ya kisi custom field ke basis par price dynamically calculate karta hoon, jaise section 6.2 wala example.

Chapter 6 Test

WOOCOMMERCE

1. Product price ko custom logic se modify karne ke liye kaunsa filter use karoge?
2. Checkout page par custom field kaise add karte ho?
3. Weight-based pricing ka logic apne alfaz mein ek line mein samjhao.
4. `woocommerce_add_cart_item_data` filter mein hamesha `$cart_item_data` return karna kyun zaroori hai?
5. `woocommerce_cart_calculate_fees` mein `is_admin()` check kyun lagate hain?
6. `woocommerce_order_status_changed` action ke teen parameters kya hain?
7. `woocommerce_checkout_process` aur `woocommerce_thankyou` mein farq kya hai (kab fire hote hain)?

Answers: Answer Key section mein (Chapter 6 – WooCommerce).

NOTES & PRACTICE SPACE

Project Explaining

Behavioral + technical mix. Interview mein "apne kisi project ke baare mein batao" zaroor poocha jayega — aur yahi wo jagah hai jahan aksar achhe developers bhi weak lagte hain, sirf isliye ke unhone structure ke bina bolna shuru kar diya.

IS CHAPTER MEIN

- Problem → Solution → Result formula
- Ek ready example answer
- Follow-up questions ki tayyari

7.1 Formula — Problem → Solution → Result

Har project ka jawab isi teen-hisson wale format mein do. Ye format interviewer ko sochne ka tareeqa dikhata hai, sirf tools ki list nahi.

1. **Problem** — client ya business ke saamne kya masla tha? (1–2 line)
2. **Solution** — tumne technically kya banaya, kaunse tools/approach use kiye aur kyun? (yahan technical depth dikhao)
3. **Result** — kya faida hua? Number ho to aur behtar (time saving, speed, sales, users).

7.2 Example Answer (Mobile App Backend Project)

SAMPLE ANSWER

"Maine ek WooCommerce store ke liye headless backend banaya jo mobile app se connect hota tha. **Problem** ye thi ke mobile app ko WordPress admin access nahi de sakte the. **Solution** ke tor par maine custom REST API endpoints banaye jisme login ke liye JWT authentication tha, cart aur checkout server-side handle hota tha, aur Stripe se payment integrate ki thi. **Result** ye hua ke app seedha WooCommerce data use kar leta tha bina WordPress admin expose kiye."

PRACTICE TIP

Isko apne alfaz mein bol kar practice karo, taake naturally nikle — ratta hua na lage. 2 minute ka target rakho: 20 second problem, 60–70 second solution, 20 second result.

7.3 Follow-up Questions Ke Liye Tayyari

Project batane ke baad interviewer usi project ke andar se technical sawal poochta hai. Har project ke liye ye 4 jawab pehle se tayyar rakho:

- **"Sabse mushkil part kya tha?"** — ek asli technical challenge aur usko solve karne ka tareeqa.
- **"Security kaise handle ki?"** — JWT / nonce / sanitize-escape (Chapter 4).
- **"Performance ka kya scene tha?"** — caching, query optimization, image optimization (Chapter 8).
- **"Ab dobara banate to kya alag karte?"** — ye sawal maturity check karta hai; ek honest improvement bata do.

APNE PROJECT KE NOTES (PROBLEM → SOLUTION → RESULT)

SEO & Performance Optimization

Agar CV ya LinkedIn headline mein "Technical SEO" aur numbers (jaise 35–40% load time reduction, PageSpeed 85+) likha hai, to interviewer wo zaroor cross-check karega. Is chapter mein wahi concepts hain jinse tum apna claim confidently defend kar sako.

IS CHAPTER MEIN

- Core Web Vitals (LCP, INP, CLS)
- Image optimization
- Page caching vs object caching
- Minification & asset combining
- Database cleanup

8.1 Core Web Vitals — Kya Hote Hain

Google ke 3 core metrics jo real user experience measure karte hain:

- **LCP (Largest Contentful Paint)** — page ka sabse bara element (usually hero image ya heading) kitni jaldi dikhta hai. **Target: 2.5 second se kam.**
- **INP (Interaction to Next Paint)** — user click/tap kare to response kitni jaldi aata hai. **Target: 200 millisecond se kam.**
- **CLS (Cumulative Layout Shift)** — page load hote waqt content kitna "jump"/shift karta hai (jaise image load hone se text neeche khisak jaye). **Target: 0.1 se kam.**

8.2 Image Optimization

```
// Lazy loading WordPress 5.5+ mein default enabled hai; force karna ho to:  
add_filter('wp_lazy_loading_enabled', '__return_true');
```

Practical steps: images ko WebP format mein convert karna, original resolution ki jagah correct display size mein resize karke upload karna, aur compression plugins (Smush, ShortPixel) use karna. Bari images site ki speed ka sabse common culprit hoti hain.

8.3 Caching

Caching se server baar baar same PHP/database query nahi chalata — pehle se generated result serve kar deta hai.


```
// Object cache example
$cached_data = wp_cache_get('my_expensive_query');

if (false === $cached_data) {
    $cached_data = expensive_database_query();
    wp_cache_set('my_expensive_query', $cached_data, '', 3600); // 1 hour
}
```

INTERVIEW QUESTION

"Page caching aur object caching mein farq?"

Answer: Page caching poori HTML page ko cache karta hai (jaise WP Rocket, W3 Total Cache) — dobara request aane par PHP chalta hi nahi. Object caching specific database query results cache karta hai (jaise Redis, Memcached) — page to generate hoti hai, lekin heavy queries dobara nahi chaltin.

8.4 Minification & Combining Assets

CSS/JS files ko minify karna (whitespace aur comments hata dena) aur combine karna, taake HTTP requests kam hon. Plugins jaise Autoptimize, ya Elementor ke apne "Improved Asset Loading" settings ye kaam automatically kar dete hain.

8.5 Database Cleanup

```
-- Post revisions clean karna (ye bohat zyada rows bana lete hain)
DELETE FROM wp_posts WHERE post_type = 'revision';

-- Spam comments clean karna
DELETE FROM wp_comments WHERE comment_approved = 'spam';
```

ZAROORI EHTIYAT

Live site par koi bhi DELETE query chalane se pehle **database ka backup** lena zaroori hai. Ek galat WHERE clause poora data uda sakta hai.

8.6 Common Interview Question

INTERVIEW QUESTION

"Tumne kisi real project ki speed kaise improve ki?"

Answer format: "Maine [image compression / caching plugin / unused plugin removal] kiya, jisse PageSpeed score [before] se [after] tak improve hua, aur load time [X]% kam hua."

CONSISTENCY CHECK

Apne real project ka actual number yahan fill karo. CV mein jo claim kiya hai, wahi story yahan bhi honi chahiye – taake tum usko genuinely defend kar sako.

Chapter 8 Test

SEO & PERFORMANCE

1. LCP, INP aur CLS kya measure karte hain?
2. Lazy loading images ke liye kaise enable karte ho?
3. Page caching aur object caching mein farq batao.
4. Post revisions database ko slow kyun karte hain?
5. Apna real speed-improvement example bol kar practice karo (numbers ke sath).

Answers: Answer Key section mein (Chapter 8 – SEO & Performance).

NOTES & PRACTICE SPACE

Elementor & Figma to WordPress

Ye tumhari core identity hai — "Elementor Specialist" aur "Figma to WordPress". Is chapter mein wahi concepts likhe hue hain jo tum daily kaam mein use karte ho, taake interview mein clearly explain kar sako.

IS CHAPTER MEIN

- Figma se WordPress workflow
- Elementor custom widget
- Elementor performance issues
- Dynamic tags (ACF + Elementor)

9.1 Figma Se WordPress Tak — Workflow

1. Figma design ko logical sections mein break karna (hero, features, testimonials, footer).
2. Har section ka spacing, typography aur colors Figma ke "Inspect" panel se exact values nikalna.
3. Elementor mein section-by-section rebuild karna, aur har responsive breakpoint (desktop, tablet, mobile) alag se design karna.
4. Images optimize karke export karna — icons ke liye SVG, photos ke liye WebP.

INTERVIEW QUESTION

"Figma design pixel-perfect WordPress mein kaise convert karte ho?"

Answer: Figma ke Inspect panel se exact spacing aur font-size values nikalta hoon, Elementor ke responsive controls se har breakpoint alag match karta hoon, aur end mein Figma design ke sath side-by-side compare karke fine-tune karta hoon.

9.2 Elementor Custom Widget Banana

Jab Elementor ke built-in widgets kaafi na hon, apna custom widget bana sakte hain:

```
class Custom_CTA_Widget extends \Elementor\Widget_Base {  
  
    public function get_name() {  
        return 'custom_cta';  
    }  
  
    public function get_title() {  
        return 'Custom CTA';  
    }  
}
```

```

protected function register_controls() {
    $this->add_control('cta_text', [
        'label' => 'Button Text',
        'type' => \Elementor\Controls_Manager::TEXT,
        'default' => 'Click Here',
    ]);
}

protected function render() {
    $settings = $this->get_settings_for_display();
    echo '<a class="cta-btn">' . esc_html($settings['cta_text']) . '</a>';
}
}

```

`\Elementor\Widget_Base` extend karke apna widget banate hain. `register_controls()` mein Elementor editor ke fields define hote hain (text, color, image waghera), aur `render()` mein actual HTML output hota hai.

9.3 Elementor Performance Issues

INTERVIEW QUESTION

"Page builders (Elementor) websites ko slow kyun kar dete hain, aur kaise fix karte ho?"

Answer: Elementor har page ke liye bohat sara CSS/JS load karta hai, chahe wo widget use ho ya na ho. **Fix:** Elementor ke "Improved CSS Loading" aur "Improved Asset Loading" experimental settings enable karna, unused widgets disable karna, aur jahan zaroorat na ho wahan native theme templates use karna.

9.4 Dynamic Tags (ACF + Elementor Integration)

Elementor Pro mein ACF fields ko dynamically show karna directly built-in "ACF Field" dynamic tag se hota hai — zyada tar cases mein koi custom code nahi chahiye. Custom dynamic tag sirf tab banate hain jab kuch special logic chahiye ho:

```

add_action('elementor/dynamic_tags/register', function($dynamic_tags) {
    // custom dynamic tag class yahan register hoti hai
});

```

1. Figma design ko pixel-perfect convert karne ka process kya hai?
2. Elementor custom widget banane ke liye kaunsi base class extend karte ho?
3. Elementor websites slow kyun ho jati hain, fix kya hai?
4. ACF field ko Elementor mein dynamically dikhane ka simplest tareeqa kya hai?

Answers: Answer Key section mein (Chapter 9 – Elementor & Figma).

NOTES & PRACTICE SPACE

Debugging & Troubleshooting

"Website fixes & troubleshooting" har client project ka hissa hai. Interviewer ye dekhna chahta hai ke tum panic karte ho ya systematically issue isolate karte ho.

IS CHAPTER MEIN

- WP_DEBUG enable karna
- Common errors ka matlab
- Debugging tools
- Plugin conflict isolate karna

10.1 WP_DEBUG Enable Karna

```
// wp-config.php mein
define('WP_DEBUG', true);
define('WP_DEBUG_LOG', true); // errors ko file mein save karna
define('WP_DEBUG_DISPLAY', false); // screen par errors na dikhana
```

Errors `wp-content/debug.log` file mein save hote hain. `WP_DEBUG_DISPLAY` ko live site par hamesha `false` rakhna chahiye, warna errors screen par dikh kar site ka internal structure leak ho sakta hai.

10.2 Common Errors aur Unka Matlab

- **White Screen of Death** — Fatal PHP error hota hai, kuch bhi dikhta nahi. Error log check karna zaroori hai. Usually plugin conflict ya memory limit exceed hona.
- **500 Internal Server Error** — `.htaccess` file corrupt ho gayi, ya plugin/theme conflict.
- **"Headers already sent"** — kisi PHP file mein `<?php` tag se pehle ya `?>` ke baad extra space/output hai.

10.3 Debugging Tools

- **Query Monitor plugin** — kaunsi database query slow chal rahi hai, kaunsa hook kis order mein fire ho raha hai, sab dikhata hai.
- **Browser DevTools** — Console tab JavaScript errors ke liye, Network tab slow-loading requests dekhne ke liye.

```
// Custom debug logging
error_log('Debug: ' . print_r($my_variable, true));
```

10.4 Plugin Conflict Isolate Karna

Systematic approach: sab plugins deactivate karo, phir ek ek karke wapas activate karo jab tak issue dobara na aa jaye. Theme check karne ke liye default WordPress theme (jaise Twenty Twenty-Four) par switch karke dekho ke masla theme mein hai ya plugin mein.

INTERVIEW QUESTION

"Client kehta hai site white screen de rahi hai — kya karoge?"

Answer: Pehle `WP_DEBUG` on karke error log check karunga taake exact error aur file pata chale. Agar log kuch na dikhaye, to systematically sab plugins deactivate karke ek ek activate karunga jab tak culprit plugin na mil jaye.

Chapter 10 Test

DEBUGGING

1. `WP_DEBUG_DISPLAY` ko live site par `false` kyun rakhte hain?
2. White Screen of Death ka common reason kya hota hai?
3. Plugin conflict isolate karne ka systematic tareeqa kya hai?
4. Query Monitor plugin kis kaam aata hai?

Answers: Answer Key section mein (Chapter 10 – Debugging).

NOTES & PRACTICE SPACE

Answer Key

Har chapter test ke jawab. Apne answers se compare karo — bilkul same lafz zaroori nahi, concept sahi hona zaroori hai.

Chapter 1 PHP CORE CONCEPTS

1. **b) Loosely** — PHP variable ka type value dekh kar khud decide karta hai.
2. `final` class ko extend hone se rokta hai — koi doosri class usse inherit nahi kar sakti.
3. "same" — kyunki `==` loose comparison hai, type check nahi karta. Agar `===` hota to "different" aata.
4. Encapsulation: data ko `private` / `protected` rakh kar sirf class ke apne methods se access karna.
Example: `$product->price` ko directly change na hone dena, sirf `setPrice()` method se.
5. Interface sirf method names batata hai (koi code nahi) aur class ko sab implement karna zaroori hota hai.
Abstract class kuch methods ka code de sakti hai aur kuch child class ke liye chhod sakti hai.

Chapter 2 MYSQL

1. INNER JOIN sirf matching rows deta hai (dono tables mein). LEFT JOIN left table ki sab rows deta hai; match na ho to NULL.
2. Primary Key: table ki har row ka unique identifier. Foreign Key: doosri table ke primary key ka reference, relation banane ke liye.
3. `SELECT * FROM orders WHERE total > 5000 ORDER BY created_at DESC;`
4. SQL Injection rokne ke liye — user input ko seedha query mein daalna dangerous hota hai.
5. SELECT fast ho jati hai, lekin INSERT/UPDATE thora slow — kyunki index bhi sath update hota hai.

1. Action kuch execute karta hai aur value return nahi karta. Filter data leta hai, modify karta hai, wapas return karta hai.
2. Taake wo post type WordPress REST API se accessible ho — mobile app ya JavaScript frontend use kar sake.
3. `register_taxonomy('brand', 'product', ['label' => 'Brand', 'hierarchical' => true]);`
4. CPT apna structured data type hota hai, apni taxonomy aur REST endpoint ke sath. Page sirf static content ke liye hota hai.
5. The Loop WordPress ka mechanism hai jisse posts fetch karke ek ek karke display kiye jate hain.
6. Pehle `functions.php` mein `register_nav_menus()` se location register karo, phir template file mein `wp_nav_menu()` se display karo. Admin mein Appearance → Menus se items assign karo.
7. Conflicts avoid karta hai (duplicate loading), dependencies manage karta hai (jQuery pehle load ho), aur version ke through cache busting hoti hai.
8. Jab client project mein purchased/parent theme ko directly edit karna risky ho — update aane par changes overwrite ho jate hain. Child theme is se bachata hai.
9. Repeater ek fixed structure ko baar baar repeat karta hai (jaise phone numbers ki list). Flexible Content har baar different layout choose karne deta hai (hero, testimonial, gallery — sab alag).
0. Jab ACF field save ho, us waqt custom logic chalane ke liye — jaise price recalculate karna ya kisi doosre system ko sync karna.

1. Security ke liye — taake koi file ko directly browser mein khol kar access na kar sake bina WordPress load kiye.

2. `rest_api_init`

3. Check karta hai ke request bhejne wala authorized hai ya nahi (jaise JWT token verify karna).

4. Code:

```
register_rest_route('myapp/v1', '/orders', [  
    'methods'      => 'POST',  
    'callback'      => 'create_order',  
    'permission_callback' => 'verify_user_token',  
]);
```

5. `do_action()` se banaya custom hook doosre plugins/themes ko us event ko listen karne deta hai. Normal function call sirf usi jagah chalta hai, extend nahi ho sakta.

6. `add_shortcode('my_tag', 'callback_function')` se register karte hain, phir editor mein `[my_tag]` likh kar use karte hain.

7. Jab data form/user se aaye to `sanitize_*` functions se clean karo. Jab data screen par print karo to `esc_html()` / `esc_url()` se escape karo, taake XSS na ho.

8. WordPress ka cron real server cron nahi hai — ye tabhi trigger hota hai jab koi visitor site par aata hai. Low-traffic site par scheduled task delay ho sakta hai.

9. Kyunki server ko kuch bhi store nahi karna padta — na session, na koi data. Sirf token ki signature verify hoti hai, isliye ise "stateless" kehte hain.

0. Kyunki signature `JWT_SECRET` (jo sirf server ke paas hai) se banti hai. Payload change karne se signature match nahi hogi, isliye tampered token turant reject ho jata hai.

Chapter 5 DIRECTORY SITES

1. Custom Post Type, Custom Taxonomy, Custom Fields.
2. Code:

```
$args = [  
    'post_type' => 'directory_listing',  
    'tax_query' => [[  
        'taxonomy' => 'listing_category',  
        'field' => 'slug',  
        'terms' => 'restaurant'  
    ]],  
    'meta_query' => [[  
        'key' => 'city',  
        'value' => 'Karachi',  
        'compare' => '='  
    ]]  
];  
  
$query = new WP_Query($args);
```

3. `meta_query` slow hoti hai kyunki `wp_postmeta` table properly indexed nahi hoti. Fix: custom database table banana ya optimized direct SQL query likhna.

Chapter 6 WOOCOMMERCE

1. `woocommerce_product_get_price`
2. `woocommerce_after_order_notes` action par `woocommerce_form_field()` se custom field add karte hain.
3. Product ka final price uske weight se multiply karke calculate hota hai — jaise 500/kg rate wala product agar 2kg ka hai to price 1000 ban jata hai.
4. Kyunki ye filter hai — filter ka rule hi ye hai ke data lo, modify karo, wapas return karo. Return na karne par poora cart item data khatam ho jayega.
5. Taake ye fee sirf frontend cart/checkout par add ho, admin panel mein dobara na add ho jaye (warna duplicate/wrong calculation ho sakti hai).
6. `$order_id`, `$old_status`, `$new_status`.
7. `woocommerce_checkout_process` order create hone se pehle chalta hai (validation ke liye).
`woocommerce_thankyou` order complete hone ke baad Thank You page par chalta hai.

Chapter 8 SEO & PERFORMANCE

1. LCP = sabse bara element kitni jaldi load hota hai. INP = user interaction ka response time. CLS = page load hote waqt content kitna shift karta hai.
2. `add_filter('wp_lazy_loading_enabled', '__return_true');` — WordPress 5.5+ mein already default enabled hai.
3. Page caching poori HTML page cache karta hai (PHP dobara chalta hi nahi). Object caching sirf specific database query results cache karta hai.
4. Har edit par ek nayi revision row `wp_posts` table mein ban jati hai, jisse table bara ho kar queries slow kar deta hai.
5. Apna real project bol kar practice karo — is ka koi fixed answer nahi.

Chapter 9 ELEMENTOR & FIGMA TO WORDPRESS

1. Figma ke Inspect panel se exact spacing/font-size nikalna, phir Elementor ke responsive controls se har breakpoint (desktop/tablet/mobile) alag match karna, aur end mein design se compare karke fine-tune karna.
2. `\Elementor\Widget_Base`
3. Elementor har page ke liye zyada CSS/JS load karta hai chahe use ho ya na ho. Fix: "Improved CSS/Asset Loading" experiments enable karna, unused widgets disable karna.
4. Elementor Pro ka built-in "ACF Field" dynamic tag directly use karna — custom code sirf special logic ke liye chahiye.

Chapter 10 DEBUGGING & TROUBLESHOOTING

1. Taake errors screen par visitors ko na dikhein aur site ka internal structure leak na ho — errors sirf log file mein jayein.
2. Fatal PHP error — usually plugin conflict ya memory limit exceed hona.
3. Sab plugins deactivate karke ek ek activate karna jab tak issue dobara na aaye; theme ko bhi default theme se test karna.
4. Kaunsi database query slow hai aur kaunsa hook kis order mein fire ho raha hai — sab dikhata hai, jisse debugging fast ho jati hai.

FINAL PRACTICE

Full Mock Test

Ye poora test bina kisi chapter ke reference ke solve karo, bilkul jaise real interview mein hoga. Apne answers likho (ya bol kar record karo), phir upar wale chapters se match karo. Target: 30 minute, bina Google ke.

Interview Simulation — 11 Questions

NO NOTES ALLOWED

1. `add_action` aur `add_filter` ka farq ek example ke sath samjhao.
2. Ek Custom Post Type "Property" banao jisme "Location" taxonomy aur "Price" custom field ho — directory website ke context mein.
3. REST API endpoint kaise secure karoge jo mobile app use kar raha ho?
4. WooCommerce mein product ki price ko uske weight ke hisaab se dynamic kaise banaoge?
5. INNER JOIN ka ek real example do jisme `products` aur `categories` tables involve hon.
6. Class `Rider` banao jo class `User` ko extend kare, aur usme ek method `assignOrder()` add karo.
7. Apne kisi ek project ko Problem → Solution → Result format mein 2 minute mein explain karo (bol kar practice karo, likh kar nahi).
8. "Page builder wale developer se tum alag kyun ho?" — iska ek confident jawab do.
9. JWT token ka signature kis se banta hai, aur koi client isko forge kyun nahi kar sakta?
10. Site ki speed improve karne ka apna real example numbers ke sath bolo.
11. Elementor slow websites ka kya reason hota hai, aur fix kya hai?

Ye sawal Chapters 1–10 se cover hote hain. Jo answer na aaye, uske chapter par wapas jao aur Chapter Test dobara solve karo.

SCORING

9–11 sahi: tum interview ke liye ready ho. **6–8 sahi:** weak chapters dobara parho, phir test repeat karo.
5 ya kam: Chapters 1, 3 aur 4 par dobara mehnat karo — foundation wahin se banti hai.

Quick Revision Checklist

Interview se ek raat pehle sirf ye page parhna kaafi hai. Har item par tick tab lagao jab tum usko bina kitaab dekhe apne alfaz mein explain kar sako.

- ☐ **PHP** — variables, arrays, functions, OOP (class, inheritance, interface)
- ☐ **MySQL** — SELECT, JOIN, GROUP BY, indexes, `$wpdb`
- ☐ **WordPress Core** — actions vs filters, CPT, taxonomy, custom fields, The Loop
- ☐ **Theme Development** — file structure, functions.php, menus, enqueue, widgets, child themes, template hierarchy
- ☐ **Plugin Development** — plugin structure, REST API, shortcodes, settings page, security, custom tables, cron jobs
- ☐ **JWT Authentication** — kya hai, kaise kaam karta hai, kyun secure hai
- ☐ **ACF** — Repeater vs Flexible Content, `acf/save_post` hook
- ☐ **Directory Sites** — CPT + taxonomy + meta_query combination
- ☐ **WooCommerce** — price, cart, checkout, order status aur thankyou hooks (definition + implementation dono)
- ☐ **SEO & Performance** — Core Web Vitals, caching, image optimization, apne real numbers
- ☐ **Elementor & Figma to WordPress** — pixel-perfect workflow, custom widgets, performance fix
- ☐ **Debugging** — WP_DEBUG, common errors, plugin conflict isolate karna
- ☐ **Projects** — apne 2-3 projects Problem → Solution → Result format mein explain karna practice karo

AAKHRI BAAT

Interview mein sab kuch yaad hona zaroori nahi. Jo aata hai wo confidently batao, aur jo nahi aata uske liye seedha keh do: "abhi tak maine wo use nahi kiya, lekin concept ye samajhta hoon aur seekh sakta hoon." Honesty + clear thinking, ratte hue jawab se hamesha behtar hoti hai.

Best of luck. — Noman Nadeem